

Chapter 3: The second law of thermodynamics

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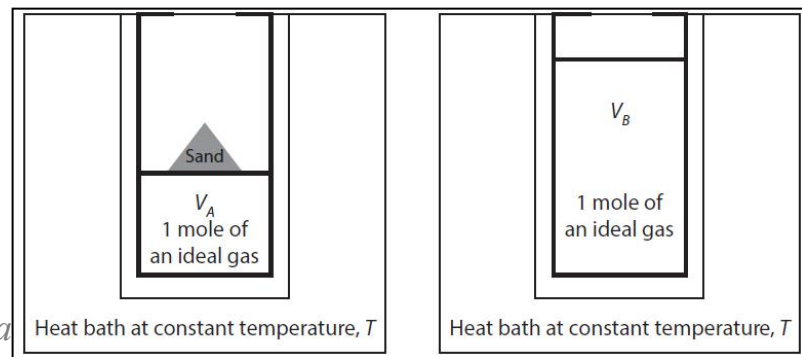
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Summary of Chapter 3 (I)

1. The processes taken by a system undergoing a change of state can be classified into two types: *reversible* and *irreversible*.
- When the change in the state of the system occurs as the result of the application of **a finite driving force**, the process proceeds irreversibly, and the degree of irreversibility of the process increases with the increasing magnitude of the driving force.
 - For a process to occur reversibly, the driving force must be **infinitesimal**, and thus, a reversible process proceeds at an infinitesimal rate. The system is imagined to move through a continuum of equilibrium states during a reversible process.



Summary of Chapter 3 (I)

2. When a system undergoes a change of state, during which it performs work and absorbs thermal energy, the magnitudes of the quantities w and q are maxima ($w_{\max}$ and q_{rev}), respectively, when the change of state occurs reversibly. For an irreversible path between the two states, less work is performed by the system, and correspondingly less thermal energy is absorbed.

3. There exists a state function called entropy, S , which can be written as

$$dS = \frac{\delta q_{\text{rev}}}{T}$$

when δq_{rev} enters (or leaves) the system. The difference between the entropy in state B and that in state A is thus

$$\Delta S = S_B - S_A = \int_A^B \frac{\delta q_{\text{rev}}}{T}$$

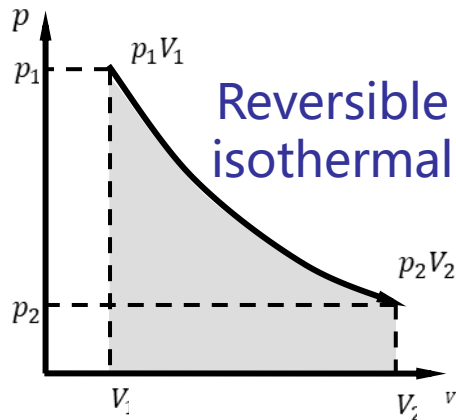
Summary of Chapter 3 (I)

4. If, in moving between the two states, the temperature of the system remains constant, the change in the entropy of the system is $\Delta S = q_{\text{rev}}/T$ where q_{rev} is the thermal energy absorbed or given off by the system in moving *reversibly* between the two states.

5. If q_{rev} is provided by a constant-temperature heat reservoir at the temperature T , the entropy of the reservoir decreases by the amount q_{rev}/T as a result of the system moving from A to B . The entropy of the combined system plus heat reservoir is thus unchanged as a result of the *reversible process*; entropy has simply been transferred from the heat reservoir to the system.

Summary of Chapter 3 (I)

Expansion

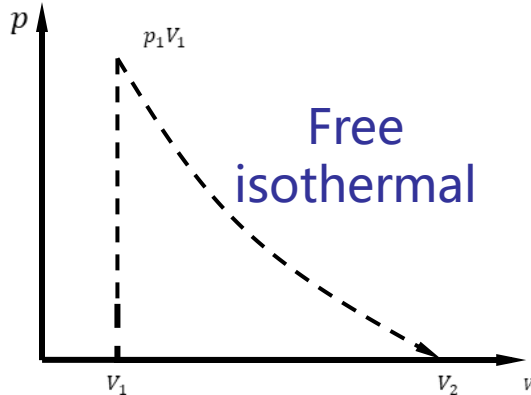


$$w_{rev} = \int_{V_A}^{V_B} P dV = \int_{V_A}^{V_B} \frac{RT}{V} dV = RT \ln \frac{V_B}{V_A}$$

$$\Delta S_{\text{gas}} = S_B - S_A = \frac{q_{rev}}{T} = \frac{w_{rev}}{T} = R \ln \frac{V_B}{V_A} > 0$$

$$\Delta S_{\text{heat reservoir}} = -\frac{q}{T} = -\Delta g_{\text{gas}} = R \ln \frac{V_A}{V_B} < 0$$

$$\Delta S_{\text{total}} = \Delta S_{\text{gas}} + \Delta S_{\text{heat reservoir}} = 0$$



$$w = \Delta U = q = 0$$

$$\Delta S_{\text{gas}} = R \ln \frac{V_B}{V_A}$$

$$\Delta S_{\text{total}} = \Delta S_{\text{gas}} + \Delta S_{\text{heat reservoir}} = \Delta S_{\text{gas}} = R \ln \frac{V_B}{V_A} > 0$$

$$0 \leq w \leq w_{\text{max}}$$

$$0 \leq q \leq q_{rev}$$

$$0 \leq \Delta S_{\text{total}} \leq \frac{q_{rev}}{T}$$

Summary of Chapter 3 (I)

6. If the changes in the state of a system from A to B were carried out *irreversibly*, then less thermal energy, q ($q < q_{\text{rev}}$), would be withdrawn from the heat reservoir by the system. Thus, the magnitude of the decrease in the entropy of the reservoir would be smaller (equal to q/T). However, since entropy is a state function, the change in entropy of the gas, $S_B - S_A$, is independent of the process path, and $\Delta S_{\text{system}} + \Delta S_{\text{heat reservoir}} > 0$. Entropy has been created as a result of the occurrence of an irreversible process. The entropy created is termed ΔS_{irr} .

7. In the general case, $S_B - S_A = q/T + \Delta S_{\text{irr}}$, and as the degree of irreversibility increases, the thermal energy, q , withdrawn from the heat reservoir decreases and the magnitude of ΔS_{irr} increases.

3.9 Summary statements

◆ The following points have emerged from the discussion so far:

1. Entropy is a thermodynamic state variable (function).
2. Entropy is not created when a system undergoes a reversible process; entropy is transferred from one part of the system/surroundings to another part.
3. The total entropy of the universe increases when an irreversible process occurs.
4. For all processes, we can write $\Delta S_{\text{system}} = q/T + \Delta S_{\text{irr}}$ and $q \leq q_{\text{rev}}$.
5. For all processes, the entropy of the universe increases or stays the same. The total entropy of the universe never decreases.

3.10 The properties of heat engines

- ◆ Traditionally the concept of entropy as a state function is introduced by considering the behavior and properties of heat engines.
- ◆ A heat engine is a device which converts heat into work, and it is interesting to note that the first steam engine, which was built in 1769, was in operation for a considerable number of years before the reverse process, i.e., the conversion of work into heat, was investigated.
- ◆ In the operation of a heat engine, some of the energy that has been transferred from a high-temperature heat reservoir is converted into work, with the remainder of the energy being transferred to a low-temperature heat reservoir.

3.10 The properties of heat engines

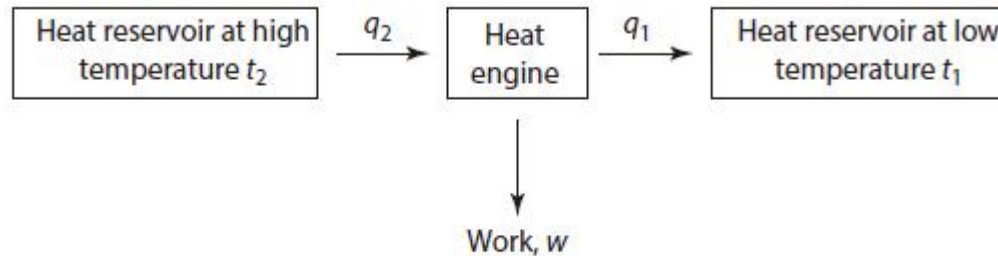


Figure 3.3 Schematic representation of the working of a heat engine.

$$\Delta U = q_2 - q_1 - w$$

- ◆ Consider the steam engine, which is a typical example of a heat engine. In a steam engine, superheated steam is passed from the boiler (the high-temperature heat reservoir) to the cylinders, where the steam performs work by expanding against the pistons (the engine).
- ◆ As a result of this expansion, the temperature of the steam decreases, and at the end of the piston stroke, the spent steam is exhausted to the atmosphere (the low-temperature heat reservoir). A flywheel returns the piston to its original position, thus completing the cycle and preparing for the next working stroke.

3.10 The properties of heat engines

- ◆ The efficiency of a heat engine is given by
- ◆ The factors governing the efficiency of this process were explained in 1824 by Sadi Carnot, who considered the cyclic process.

$$\text{Efficiency} = \frac{\text{work obtained}}{\text{heat input}} = \frac{w}{q_2}$$

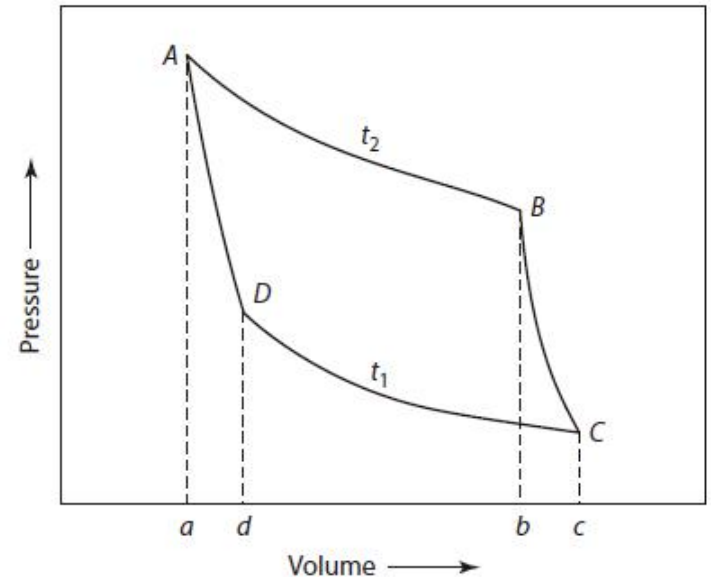


Figure 3.4 A pressure vs. volume depiction of a Carnot cycle.

3.10 The properties of heat engines

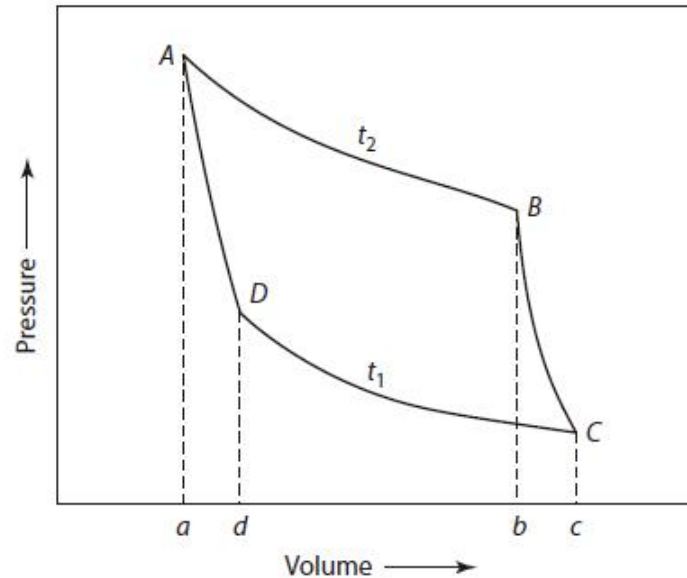


Figure 3.4 A pressure vs. volume depiction of a Carnot cycle.

- In the step **A→B**, thermal energy q_2 is isothermally and reversibly transferred from a heat reservoir at the temperature t_2 to a thermodynamic substance, as a result of which the thermodynamic substance **isothermally and reversibly expands** from the state A to the state B and performs work w_1 equal to the area $ABba$.

3.10 The properties of heat engines

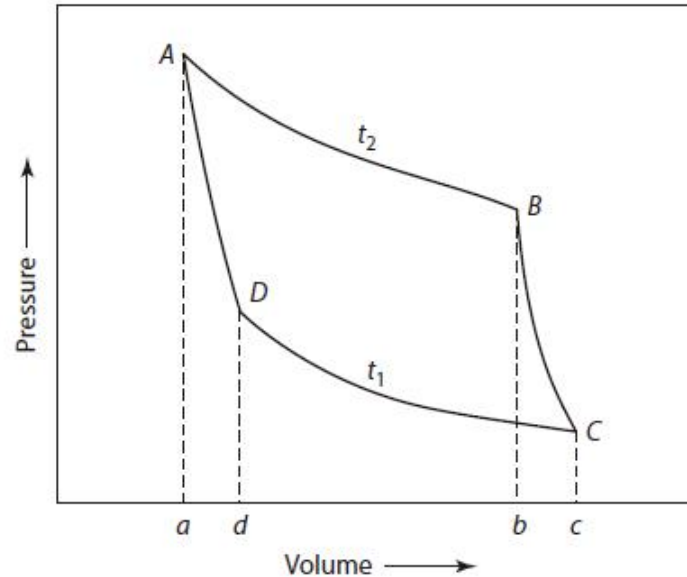


Figure 3.4 A pressure vs. volume depiction of a Carnot cycle.

- In the step **B→C**, the thermodynamic substance undergoes a **reversible adiabatic expansion** from the state **B** to the state **C**, as a result of which its temperature decreases to t_1 , and it performs work equal to the area **BCcb**.

3.10 The properties of heat engines

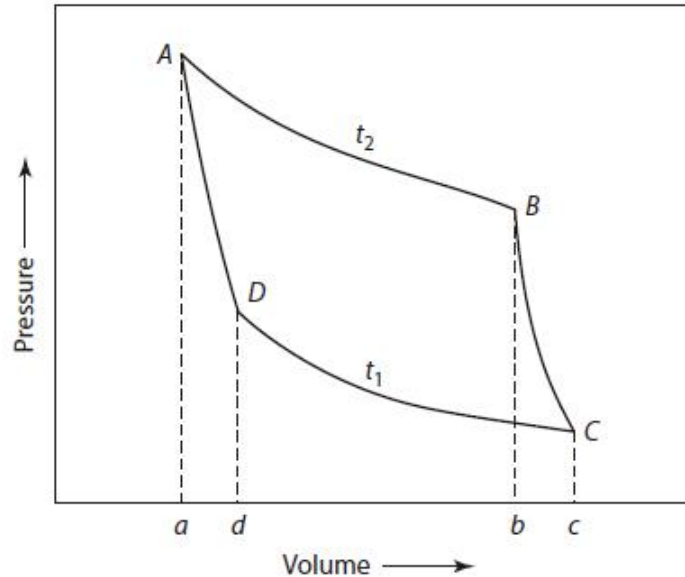
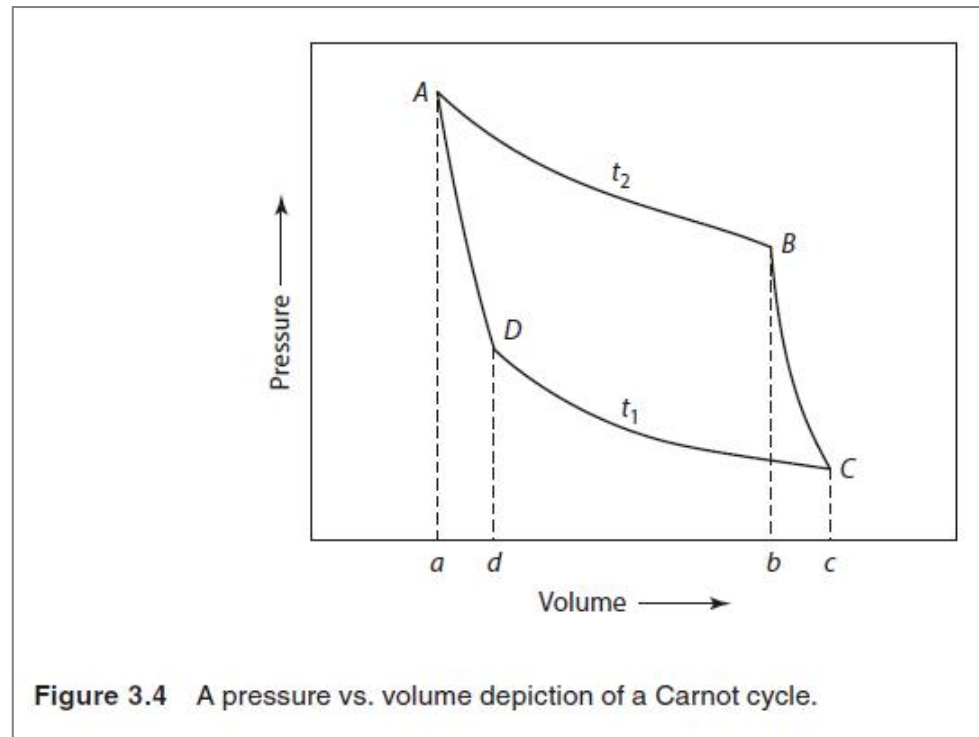


Figure 3.4 A pressure vs. volume depiction of a Carnot cycle.

- In the step **C**→**D**, energy q_1 is **isothermally and reversibly** transferred from the thermodynamic substance to a heat reservoir at the temperature t_1 . Work w_3 , equal to the area $DCcd$, is done on the substance.

3.10 The properties of heat engines



- In the step $D \rightarrow A$, the substance is reversibly and adiabatically compressed, during which its temperature increases from t_1 to t_2 , and work w_4 , equal to the area $ADda$, is done on the substance.

3.10 The properties of heat engines

During the cyclic process, which has returned the thermodynamic substance to its initial state, the substance has performed the work $w=w_1+w_2-w_3-w_4$ (equal to the area $ABCD$) and has absorbed heat $q=q_2-q_1$. For a cyclic process, $\Delta U=0$, and thus, from the First Law,

$$\sum_i q_i = \sum_i w_i \quad q = w$$
$$q_2 - q_1 = \sum_i w_i = w$$

The efficiency of the cyclic process (which is known as a Carnot cycle) is given by:

$$\text{Efficiency} = \frac{w}{q_2} = \frac{q_2 - q_1}{q_2} = 1 - \frac{q_1}{q_2}$$

This equation shows that in this idealized Carnot cycle (all processes assumed to be reversible), the efficiency is less than unity for finite values of q_1 .

3.10 The properties of heat engines

- ◆ Consider a second engine working with a different substance, again between the temperatures t_1 and t_2 , and let this second engine be more efficient than the first one. This greater efficiency could be obtained in either of two ways.
 1. q_2 is withdrawn from the heat reservoir at t_2 , and more work, w' , is obtained from it than was obtained from the first engine, i.e., $w' > w$. Thus, the second engine rejects less heat q_1' , to the cold reservoir at t_1 than does the first engine, i.e., $q' < q$.
 2. The same work is obtained by withdrawing less heat q_2' from the heat reservoir at t_2 . i.e., $q_2' < q_2$. Thus less heat, q_1' , is rejected into the heat reservoir at t_1 , $q_1' < q_1$.

3.10 The properties of heat engines

- ◆ Consider now, that the second engine is run in the forward direction, and the first engine is run in the reverse direction, i.e., acts as a heat pump.
- From case 1, for the second engine run in the forward direction, $w' = q_2 - q_1'$, for the first engine run in the reverse direction, $-w = -q_2 + q_1$, and the sum of the two processes is

$$w' - w = q_1 - q_1' < 0$$

i.e., an amount of work $(w' - w)$ has been obtained from a quantity of thermal energy $(q_1 - q_1')$ without any other change occurring. Although this conclusion does not contravene the First Law of Thermodynamics, it is contrary to human experience. Such a process corresponds to **perpetual motion of the second kind**; that is, heat is converted to work without leaving a change in any other body. (**Perpetual motion of the first kind** is the creation of energy from nothing)

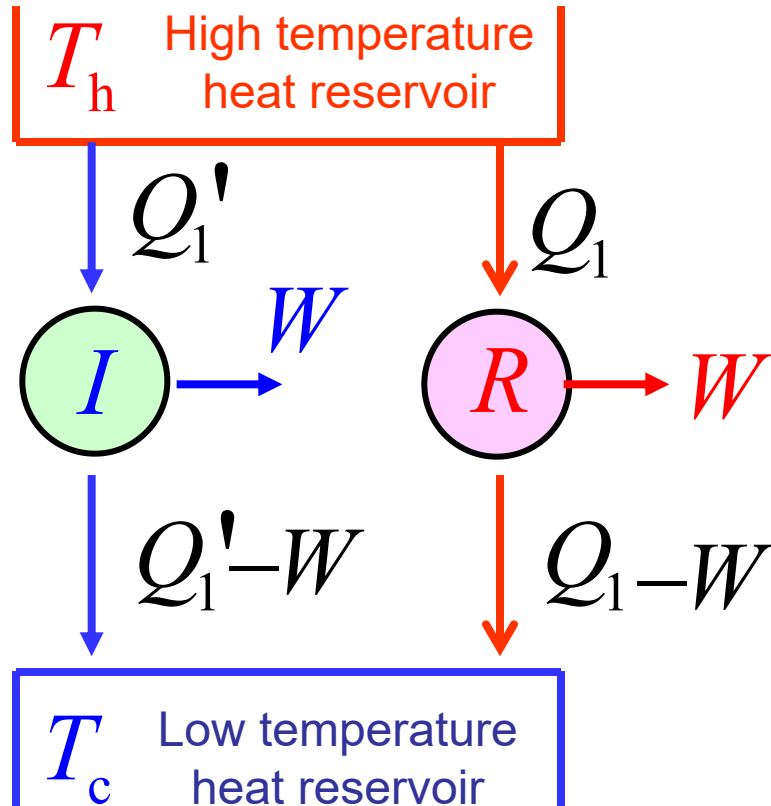
3.10 The properties of heat engines

- From case 2, for the second engine run in the forward direction, $w = q'_2 - q'_1$. For the first engine run in the reverse direction, $-w = -q_2 + q_1$. The sum of the two processes is

$$q_2 - q'_2 = q_1 - q'_1 = q$$

that is, an amount of thermal energy at the lower temperature has been transferred to a higher temperature without any other change occurring. This corresponds to the spontaneous transfer of thermal energy up a temperature gradient and is thus even more contrary to human experience.

3.10 The properties of heat engines



(a)

$$W = W$$

$$\eta_I = \frac{W}{Q_1'}$$

$$\eta_R = \frac{W}{Q_1}$$

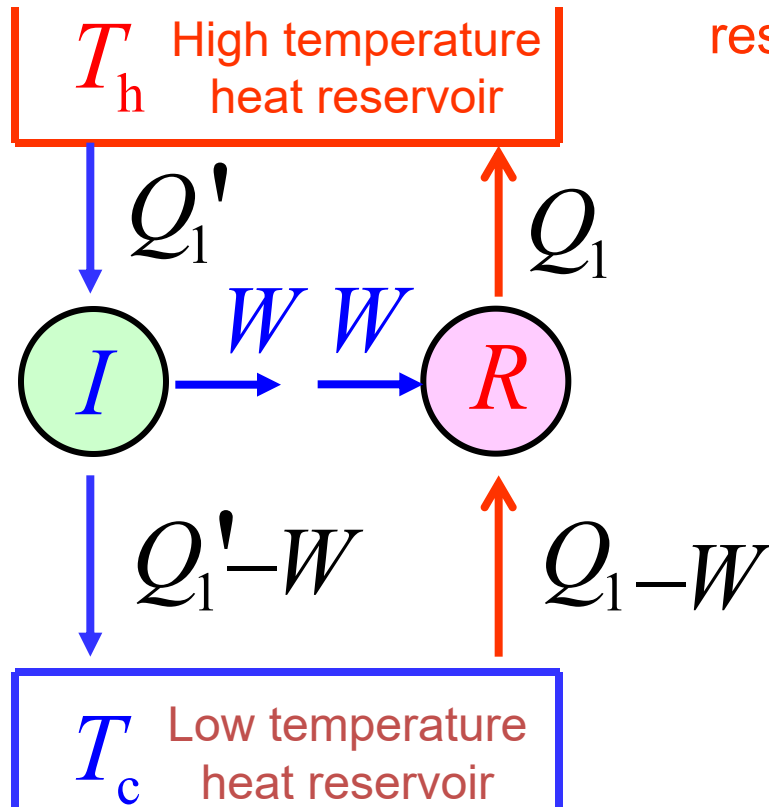
Assuming:

$$\eta_I > \eta_R$$

$$\frac{W}{Q_1'} > \frac{W}{Q_1}$$

$$Q_1 > Q_1'$$

3.10 The properties of heat engines



(b)

Absorb heat from low temperature heat reservoir

$$(Q_1 - W) - (Q_1' - W) \\ = (Q_1 - Q_1') > 0$$

High temperature heat reservoir gets heat

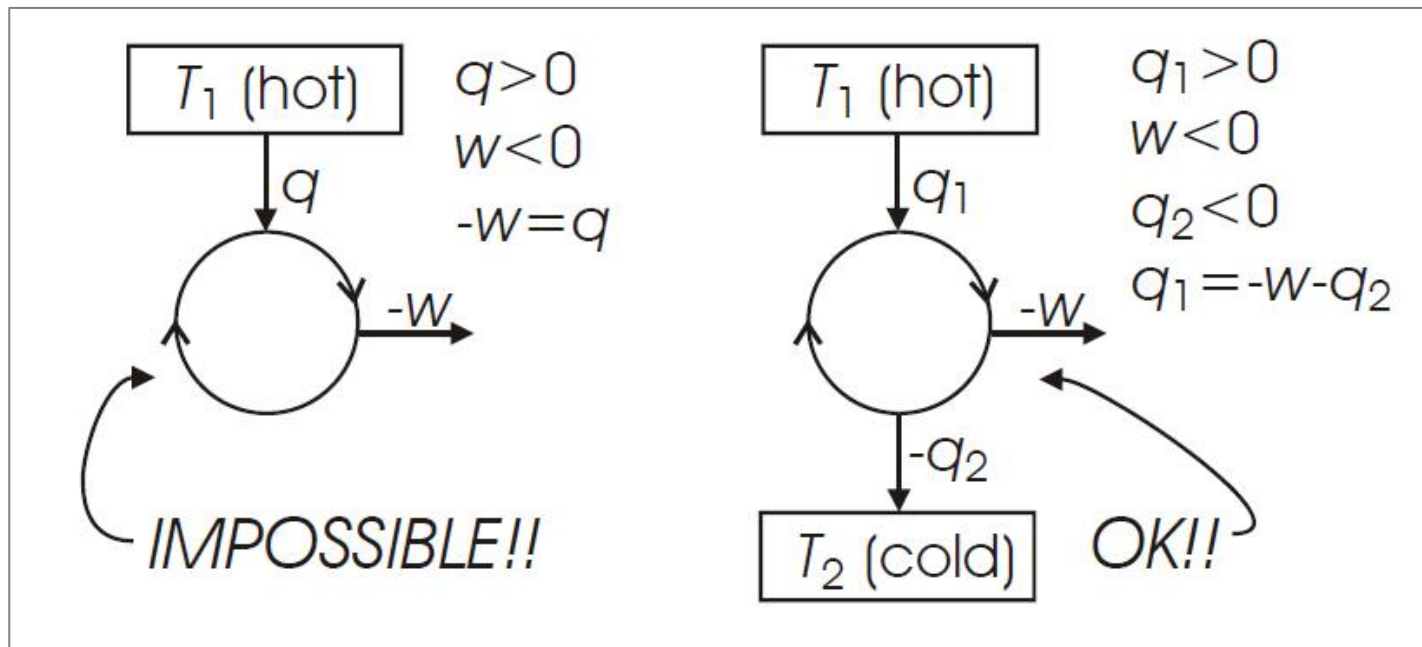
$$(Q_1 - Q_1') > 0$$

This violates Clausius's statement.

$$\eta_I \leq \eta_R$$

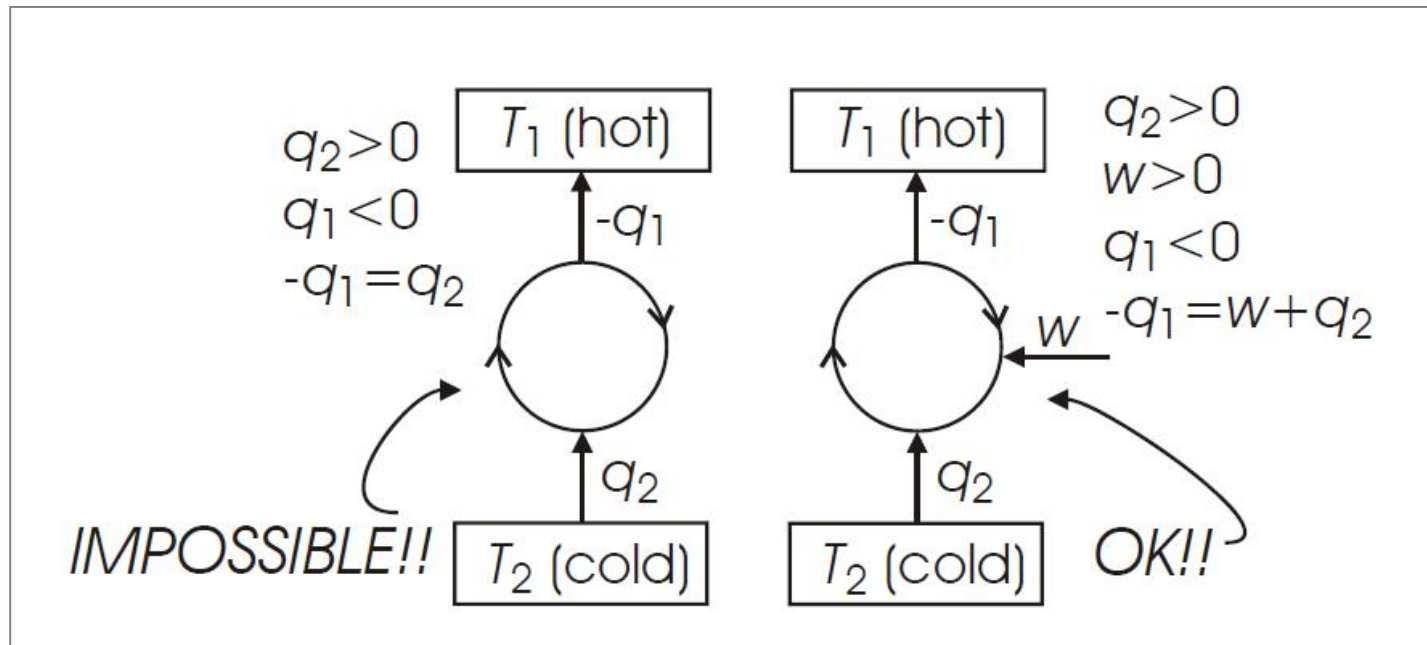
3.10 The properties of heat engines

- ◆ The preceding discussion gives rise to the following two preliminary formulations of the Second Law of Thermodynamics:
 - The principle of Thomsen states that it is impossible, by means of a cyclic process, to take heat from a reservoir and convert it to work without, in the same operation, transferring heat to a cold reservoir.



3.10 The properties of heat engines

- ◆ The preceding discussion gives rise to the following two preliminary formulations of the Second Law of Thermodynamics:
 - The principle of Clausius states that it is impossible to transfer heat from a cold to a hot reservoir without, in the same process, converting a certain amount of work to heat.



3.11 The thermodynamic temperature scale

- ◆ All reversible Carnot cycles operating between the same upper and lower temperatures must have the same efficiency—namely, the maximum possible. This maximum efficiency is independent of the working substance and is a function only of the working temperatures t_1 and t_2 . Thus,

$$\text{Efficiency} = \frac{q_2 - q_1}{q_2} = 1 - \frac{q_1}{q_2} = f'(t_1, t_2)$$

$$\text{or } \frac{q_1}{q_2} = f(t_1, t_2)$$

3.11 The thermodynamic temperature scale

$$\text{Efficiency} = \frac{q_2 - q_1}{q_2} = 1 - \frac{q_1}{q_2} = f'(t_1, t_2)$$

$$\text{or } \frac{q_1}{q_2} = f(t_1, t_2)$$

$$\frac{q_1}{q_3} \times \frac{q_3}{q_2} = \frac{f(t_1, t_3)}{f(t_2, t_3)} = \frac{q_1}{q_2} = f(t_1, t_2)$$

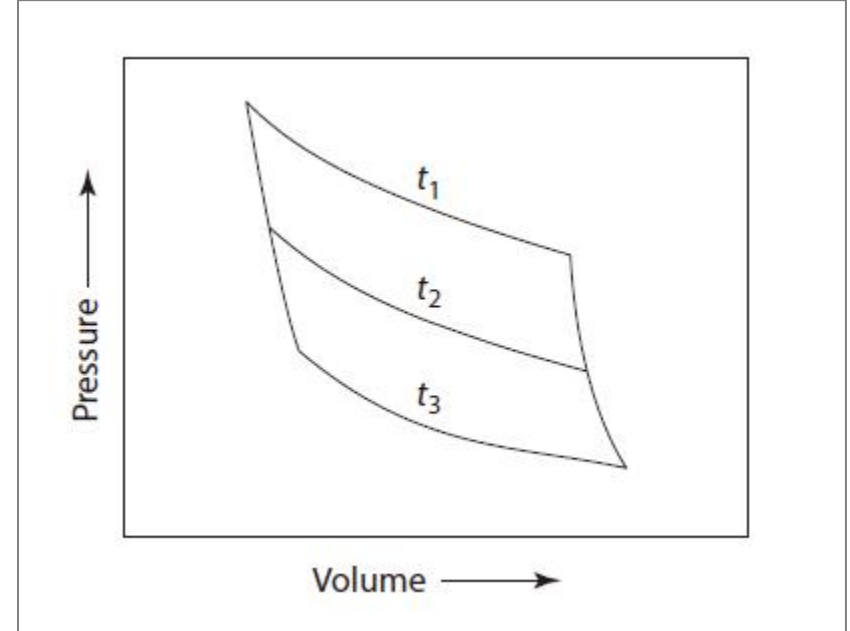


Figure 3.5 Carnot cycles operating between t_1 and t_2 , t_1 and t_3 , and t_2 and t_3 .

- ✓ As $f(t_1, t_2)$ is independent of t_3 , then $f(t_1, t_3)$ and $f(t_2, t_3)$ must be of the form $f(t_1, t_3) = F(t_1)/F(t_3)$ and $f(t_2, t_3) = F(t_2)/F(t_3)$, i.e., the efficiency function $f(t_1, t_2)$ is the quotient of a function of t_1 alone and t_2 alone.

$$\frac{q_1}{q_2} = \frac{F(t_1)}{F(t_2)}$$

3.11 The thermodynamic temperature scale

- ◆ Kelvin took these functions to have the simplest possible form, namely, T_1 and T_2 . Thus

$$\frac{q_1}{q_2} = \frac{F(t_1)}{F(t_2)} = \frac{T_1}{T_2}$$

In this case, the efficiency of a Carnot cycle is:

$$\text{Efficiency} = \frac{q_2 - q_1}{q_2} = \frac{T_2 - T_1}{T_2} = 1 - \frac{T_1}{T_2}$$

- This defines an absolute thermodynamic scale of temperature which is independent of the working substance. It is seen that the zero of this temperature scale is that temperature of the cold reservoir which makes the Carnot cycle 100% efficient.

3.11 The thermodynamic temperature scale

- ◆ The absolute thermodynamic temperature scale (or Kelvin scale) is identical with the ideal gas temperature scale. This can be demonstrated by considering 1 mole of ideal gas to be the working substance in a Carnot cycle.

- State A to state B. Reversible isothermal expansion at t_2 :

$$\Delta U = 0, \quad q_2 = w_1 = R t_2 \ln \frac{V_B}{V_A}$$

- State B to state C. Reversible adiabatic expansion:

$$q = 0$$

$$w_2 = -\Delta U = -\int_{t_2}^{t_1} c_V dT$$

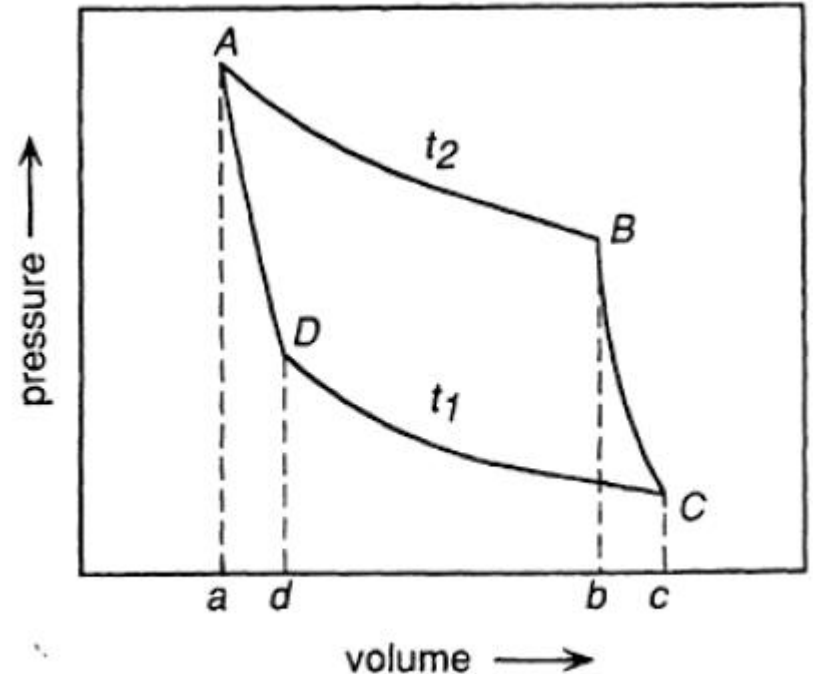


Figure 3.4 The Carnot cycle.

3.11 The thermodynamic temperature scale

- State C to state D. Reversible isothermal compression at t_1 :

$$\Delta U = 0$$

$$q_1 = w_3 = Rt_1 \ln \frac{V_D}{V_C}$$

- State D to state A. Reversible adiabatic compression:

$$q = 0$$

$$w_t = -\Delta U = -\int_{t_1}^{t_2} c_V dT$$

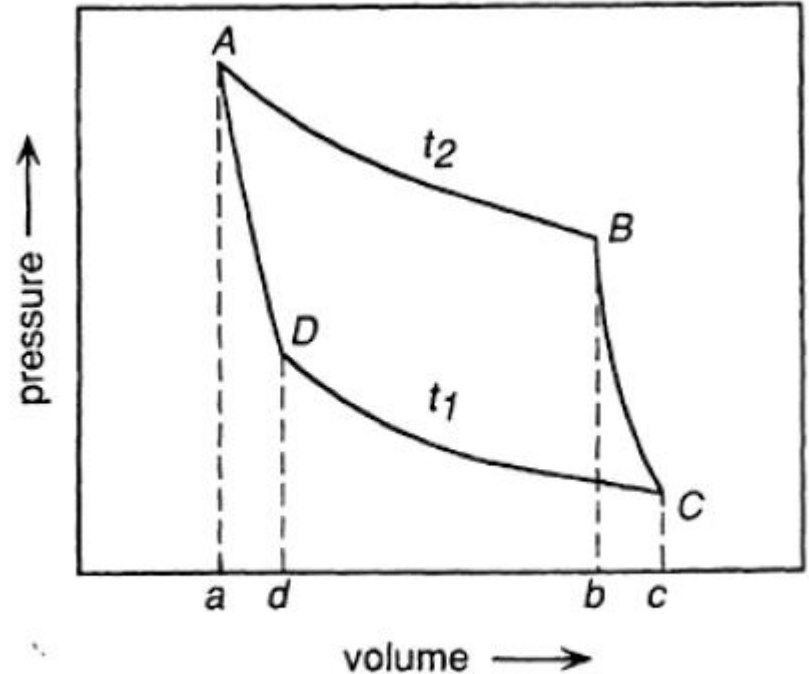


Figure 3.4 The Carnot cycle.

3.11 The thermodynamic temperature scale

◆ Total work done on the gas

$$\begin{aligned} w &= w_1 + w_2 + w_3 + w_4 \\ &= R t_2 \ln \frac{V_B}{V_A} - \int_{t_2}^{t_1} c_V dT + R t_1 \ln \frac{V_D}{V_C} - \int_{t_1}^{t_2} c_V dT \end{aligned}$$

$$\frac{V_B}{V_A} = \frac{V_C}{V_D} \quad \Rightarrow \quad w = R(t_2 - t_1) \ln \frac{V_B}{V_A}$$

◆ The heat absorbed from the hot reservoir

$$q_2 = R t_2 \ln \frac{V_B}{V_A} \quad \text{Efficiency} = \eta = \frac{w}{q_2} = \frac{t_2 - t_1}{t_2} = 1 - \frac{T_1}{T_2}$$

3.12 The second law of thermodynamics

$$1 - \frac{T_1}{T_2} = 1 - \frac{q_1}{q_2}$$

$$\frac{q_2}{T_2} - \frac{q_1}{T_1} = 0$$

- ◆ Now, any cyclic process can be broken down into a number of Carnot cycles. In going round the cycle ABA in a clockwise direction, the work done by the system equals the area enclosed by the path loop. This loop can be roughly approximated by a number of Carnot cycles as shown, and for the zigzag paths of these cycles.

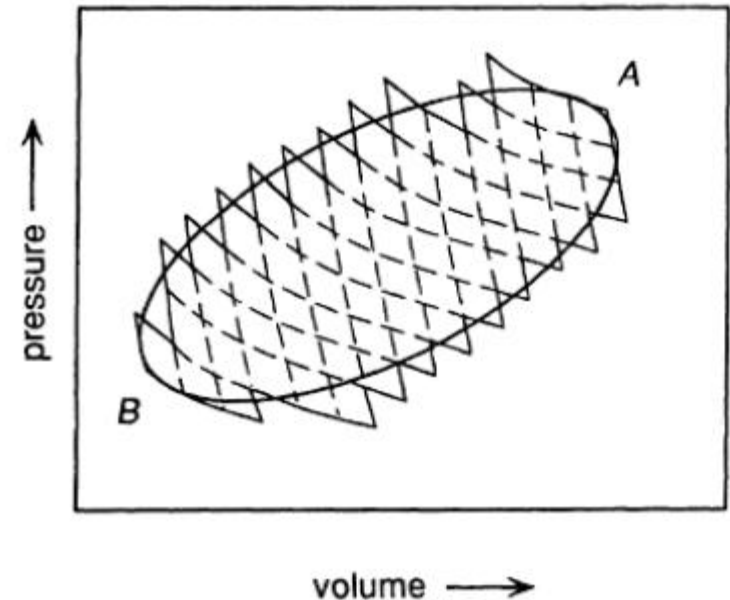


Figure 3.6 A cyclic process broken down into a large number of Carnot cycles.

3.12 The second law of thermodynamics

$$\sum \frac{q_i}{T_i} = 0$$

- ◆ The heat entering the system is positive and the heat leaving the system is negative. The zigzag path of the Carnot cycles can be made to coincide with loop ABA by making the Carnot cycles smaller and smaller, and in the limit of coincidence, the summation can be replaced by a cyclic integral:

$$\oint \left(\frac{\delta q_i}{T} \right)_{rev} = 0$$

- ◆ The vanishing of the cyclic integral indicates that the integral is a perfect differential of some function of state of the system. This function is called the entropy, S , and is defined as

$$dS = \frac{\delta q}{T}$$

3.12 The second law of thermodynamics

$$dS = \frac{\delta q}{T}$$

- ◆ It is to be emphasized that q is the reversible heat increment, and thus the equation should be properly written as:

$$dS = \frac{\delta q_{\text{rev}}}{T}$$

- ◆ The application to a reversibly operated heat engine, in which q_2 is withdrawn from a constant-temperature source at T_2 , work w is performed, and heat q_1 is rejected into a constant-temperature heat sink at T_1 , shows that the decrease in the entropy of the heat source, q_2/T_2 , equals the increase in the entropy of the heat sink, q_1/T_1 , i.e., $\Delta S_{\text{total}}=0$, which is a consequence of the fact that the process is conducted reversibly.

3.12 The second law of thermodynamics

- ◆ The Second Law of Thermodynamics can thus be stated as
 - The incremental change in entropy dS of a system into which (or out of which) thermal energy has been reversibly added (or expelled) is given as $dS = \delta q_{\text{rev}}/T$, and the function S is a state function of the system.
 - The entropy of a system in an adiabatic enclosure can never decrease.
 - The entropy increases during an irreversible process.
 - The entropy remains constant during a reversible process.
 - The entropy remains constant if the system was initially in equilibrium.

3.12 The second law of thermodynamics

- ◆ For an infinitesimal change of state of an adiabatically contained system,

$$dS_i \geq 0$$

that is, the sum of the incremental changes in entropy of all i parts of the system which are in thermal contact with each other is zero if the infinitesimal change of state is reversible and is greater than zero if the infinitesimal change of state is irreversible. It can be converted to an equality by writing

$$\sum dS_i = dS_{irr}$$

where dS_{irr} is the total entropy created in the given incremental irreversible process.

3.13 Maximum work

- ◆ For a change of state from A to B, the First Law gives

$$U_B - U_A = q - w$$

The First Law gives no indication of the allowed magnitudes of q and w in the given process. Although the values of q and w can vary depending on the degree of irreversibility of the path taken between the state A and B, the Second Law sets a definite limit on the maximum amount of work which can be obtained from the system during a given change of state, and, hence, sets a limit on the quantity of heat which the system may absorb. For an infinitesimal change of state,

$$dS_{\text{system}} = \frac{\delta q}{T} + dS_{\text{irr}}$$

3.13 Maximum work

The Second Law: $dS_{\text{system}} = \frac{\delta q}{T} + dS_{\text{irr}}$

The First Law: $\delta q = dU_{\text{system}} + \delta w$

$$\left. \begin{array}{l} dS_{\text{system}} = \frac{\delta q}{T} + dS_{\text{irr}} \\ \delta q = dU_{\text{system}} + \delta w \end{array} \right\} dS_{\text{system}} = \frac{dU_{\text{system}} + \delta w}{T} + dS_{\text{irr}}$$

- If the temperature remains constant throughout the process (and equal to the temperature of the reservoir supplying heat to the system), then integration from state A to state B gives

$$\delta w = TdS_{\text{system}} - dU_{\text{system}} - TdS_{\text{irr}}$$

$$\delta w \leq TdS_{\text{system}} - dU_{\text{system}}$$

$$w_{\text{max}} = q_{\text{rev}} = T(S_B - S_A) - (U_B - U_A) \quad \leftarrow w \leq T(S_B - S_A) - (U_B - U_A)$$

3.13 Maximum work

- ◆ As entropy is a state function, then in undergoing any specific change of state from A to B, the change in the entropy of the system is the same whether the process is conducted reversibly or irreversibly.
- ◆ It is the heat effect which is different in the two cases, i.e., if the process involves the absorption of heat and is conducted reversibly, then the heat absorbed, q_{rev} , is greater than the heat which would have been absorbed if the process had been conducted irreversibly.
- When 1 mole of an ideal gas is isothermally and reversibly expanded from state A to state B, heat $q=RT\ln(V_B/V_A)$ is reversibly transferred from the heat reservoir to the gas, and the increase in the entropy of the gas, S_B-S_A , equals $R\ln(V_B/V_A)$. The entropy of the heat reservoir decreases by an equal amount and entropy is not created, i.e., $\Delta S_{irr}=0$.

3.13 Maximum work

- If the mole of gas is allowed to expand freely from P_A to P_B then, as the gas performs no work, no heat is transferred from the reservoir to the gas, and there is no change in the entropy of the reservoir.
- As entropy is a state function, the value of $S_B - S_A$ is independent of the process path, and hence the entropy created, ΔS_{irr} , equals $S_B - S_A = R \ln(V_B/V_A)$. This entropy is created as a result of the degradation of the work which would have been performed by the gas, which had the expansion not been carried out against a zero pressure.

This degraded work

$$\begin{aligned} &= w_{\text{max}} - w \\ &= w_{\text{max}} - 0 \\ &= w_{\text{max}} \\ &= q_{\text{rev}} \end{aligned}$$

3.13 Maximum work

- ◆ The free expansion represents the limit of irreversibility at which all of the “potential” work is degraded to heat. The degraded heat appearing in the gas accounts for the increase in the entropy of the gas. Thus, for the isothermal expansion of 1 mole of ideal gas from the state A to the state B, the value of ΔS_{irr} can vary between zero and $R \ln(V_B/V_A)$ depending on the degree of irreversibility of the process.

$$0 \leq \Delta S_{irr} \leq R \ln \frac{V_B}{V_A}$$

3.14 Entropy and the criterion for equilibrium

- ◆ It was stated that a system, left to itself, would either remain in the state in which it happened to be, or would *spontaneously* move to some other state, i.e., if the system is initially at equilibrium, then it will remain at equilibrium, and if it is initially in a non-equilibrium state then it will spontaneously move to its equilibrium state.
- ◆ This *spontaneous* process is, by definition, irreversible, and during the movement of the system from its initial non-equilibrium state to its final equilibrium state the entropy of the system increases. The attainment of the equilibrium state coincides with the entropy reaching a *maximum value*, and hence entropy can be used as a criterion for determination of the equilibrium state.

3.14 Entropy and the criterion for equilibrium

- In an *isolated* system of constant internal energy, U , and constant volume, V , equilibrium is attained when the entropy of the system is a maximum, consistent with the fixed values of U and V .
- Consider the chemical reaction $A+B=C+D$ occurring in an adiabatic enclosure at constant volume. Starting with A and B , the reaction will proceed from left to right as long as the entropy of the system is thereby increased; or, conversely, starting with C and D , the reaction will proceed from right to left, again provided that the entropy of the system is thereby increased.

3.14 Entropy and the criterion for equilibrium

Fig. 3.7 shows a possible variation of entropy with extent of reaction. It is seen that a point is reached along the reaction coordinate at which the entropy of the system has its maximum value. This is the equilibrium state of the system, as further reaction, in either direction, would decrease the entropy and, hence, will not occur spontaneously.

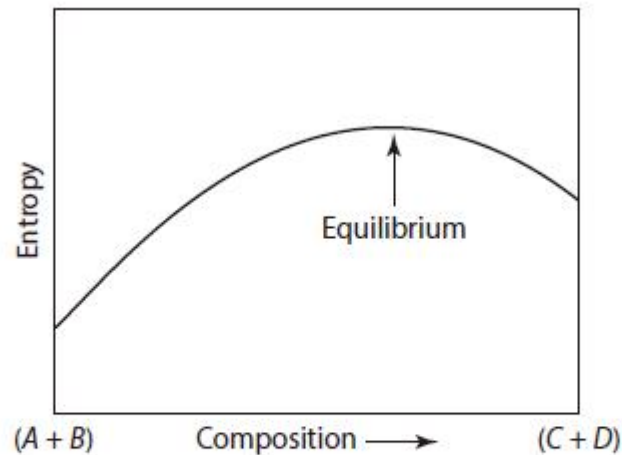


Figure 3.7 Schematic representation of the entropy of a closed system containing $A + B + C + D$ as a function of the extent of the reaction $A + B = C + D$ at constant internal energy and volume.

3.15 The combined statement of the first and second laws of thermodynamics

- ◆ For an incremental change in the state of a closed system, the First Law of Thermodynamics gives:

$$dU = \delta q - \delta w$$

- ◆ If the process occurs reversibly, the Second Law of Thermodynamics gives:

$$dS = \frac{\delta q}{T} \text{ or } \delta q = TdS \qquad \delta w = PdV$$

- ◆ Combination of the two laws gives the equation

$$dU = TdS - PdV$$

3.15 The combined statement of the first and second laws of thermodynamics

$$dU = TdS - PdV$$

◆ Restrictions on the application are

- That the system is closed, i.e., does not exchange matter with its surroundings during the process.
- That work due to change in volume is the only form of work performed by the system.

◆ This equation relates the dependent variable of the system, U , to the independent variables, S and V , i.e.,

$$U(S, V) \Rightarrow dU = \left(\frac{\partial U}{\partial S} \right)_V dS + \left(\frac{\partial U}{\partial V} \right)_S dV$$

}

$\left(\frac{\partial U}{\partial S} \right)_V = T$

$\left(\frac{\partial U}{\partial V} \right)_S = -P$

$dU = TdS - PdV$

3.15 The combined statement of the first and second laws of thermodynamics

$$dU = \left(\frac{\partial U}{\partial S} \right)_V dS + \left(\frac{\partial U}{\partial V} \right)_S dV$$

- ✓ The particularly simple form stems from the fact that, in considering variations in U as the dependent variable, the “natural” choice of independent variables is S and V .
- ✓ Consideration of S as the dependent variable and U and V as the independent variables, i.e.,

$$dS = \left(\frac{\partial S}{\partial U} \right)_V dU + \left(\frac{\partial S}{\partial V} \right)_U dV$$

3.15 The combined statement of the first and second laws of thermodynamics

$$dS = \left(\frac{\partial S}{\partial U} \right)_V dU + \left(\frac{\partial S}{\partial V} \right)_U dV \quad dS = \frac{dU}{T} + \frac{PdV}{T}$$

$$\left(\frac{\partial S}{\partial U} \right)_V = \frac{1}{T} \quad \left(\frac{\partial S}{\partial V} \right)_U = \frac{P}{T}$$

- It can be seen that increasing the internal energy of the system at constant volume increases its entropy, since $1/T > 0$.
- The curvature of the S versus U plot is negative.

$$\left(\frac{\partial^2 S}{\partial U^2} \right)_V = -\frac{1}{T^2} \frac{\partial T}{\partial U} < 0$$

3.15 The combined statement of the first and second laws of thermodynamics

$$\left(\frac{\partial S}{\partial U}\right)_V = \frac{1}{T}$$

$$\left(\frac{\partial^2 S}{\partial U^2}\right)_V = -\frac{1}{T^2} \frac{\partial T}{\partial U} < 0$$

$$\left(\frac{\partial S}{\partial V}\right)_U = \frac{P}{T}$$

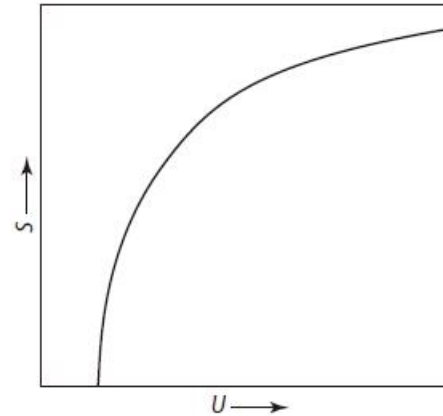


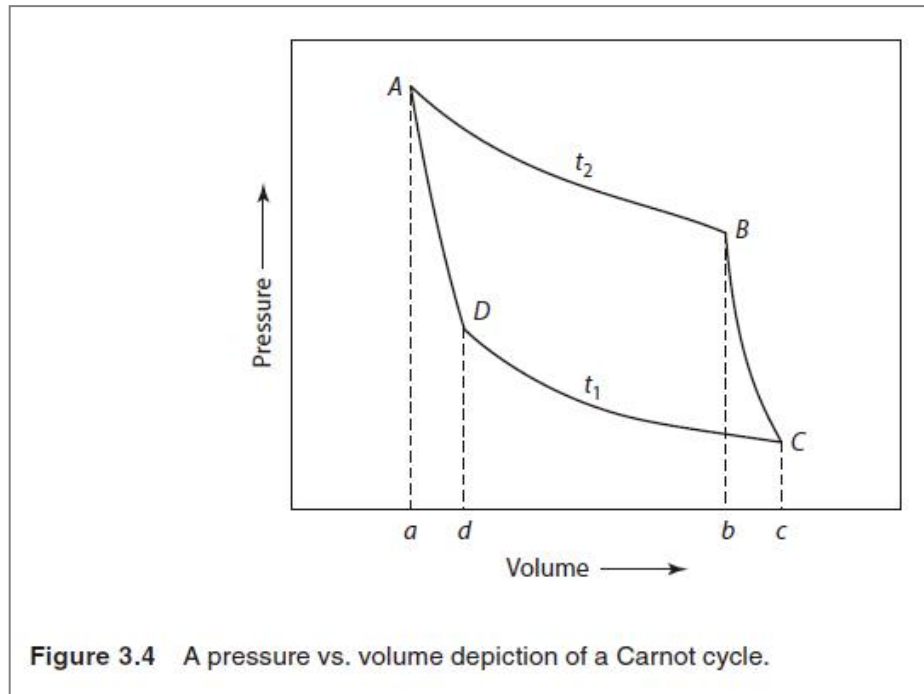
Figure 3.8 Plot of entropy vs. internal energy. Note that as U approaches its minimum value, S also approaches its minimum value.

Since $P/T > 0$, increasing the volume increases the entropy of a system at constant U . The plot of S versus V is similar to the one of S versus U : S increases with V and the plot has negative curvature.

3.18 Qualitative Example Problems

Qualitative Problem 1

A heat engine is operating in a Carnot cycle: all processes are reversible. For each of the processes, obtain expressions for the change in entropy of the engine and the change in entropy of the surroundings after one cycle. Assume one mole of an ideal monatomic gas.



3.18 Qualitative Example Problems

Solution to Qualitative Problem 1

The engine :

1. For the reversible isothermal expansion process from A to B :

$$\Delta S_{engine} = \frac{q_2}{T_2} = R \ln \frac{V_B}{V_A}$$

2. For the reversible adiabatic expansion process from B to C :

$$q = 0 \Rightarrow \Delta S_{engine} = 0$$

3. For the reversible isothermal compression process from C to D :

$$\Delta S_{engine} = \frac{q_1}{T_1} = R \ln \frac{V_D}{V_C}$$

4. For the reversible adiabatic compression process from D to A :

$$q = 0 \Rightarrow \Delta S_{engine} = 0$$

3.18 Qualitative Example Problems

The total change in entropy of the engine is therefore

$$\Delta S_{engine}^{total} = R \ln \frac{V_B}{V_A} + R \ln \frac{V_D}{V_C} = 0$$

The surroundings :

For each of the processes, the change in entropy of the surroundings is the negative of that for the engine.

$$\Delta S_{surrounding}^{total} = 0$$

$$\left. \begin{array}{l} \Delta S_{engine}^{total} = 0 \\ \Delta S_{surrounding}^{total} = 0 \end{array} \right\} \Delta S^{total} = 0 \quad \text{For the Carnot cycle}$$

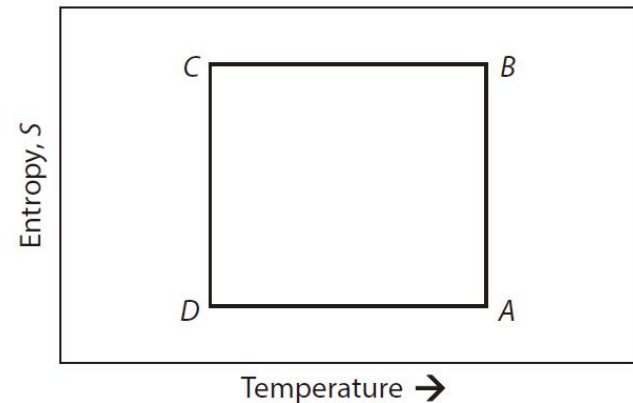
3.18 Qualitative Example Problems

Qualitative Problem 2

Draw the entropy versus temperature diagram for the Carnot cycle.

Solution to Qualitative Problem 2

1. The first process is an isothermal ($dT = 0$) increase in entropy from state A to B .
2. The second process is a constant entropy decrease in temperature from state B to C .
3. The third process is an isothermal decrease in entropy from state C to state D .
4. The last process is a constant entropy increase in temperature from state D to state A .



3.19 Quantitative Example Problems

Quantitative Problem 1

- ✓ Five moles of a monatomic ideal gas are contained adiabatically at 50 atm pressure and 300 K. The pressure is suddenly released to 10 atm, and the gas undergoes an irreversible expansion during which it performs 4000 joules of work.
- Show that the final temperature of the gas after the irreversible expansion is greater than that which the gas would attain if the expansion from 50 to 10 atm had been conducted reversibly.
- Calculate the entropy produced as a result of the irreversible expansion. The constant-volume molar heat capacity of the gas, C_v , has the value $1.5R$.

3.19 Quantitative Example Problems

□ In the initial state 1,

$$V_1 = \frac{nRT_1}{P_1} = \frac{5 \times 0.08206 \times 300}{50} = 2.46 \text{ liters}$$

□ If the adiabatic expansion from 50 to 10 atm is carried out reversibly, then the process path follows $PV^\gamma = \text{constant}$, and in the final state 2,

$$V_2 = \left(\frac{P_1 V_1^\gamma}{P_2} \right)^{1/\gamma} = \left(\frac{50 \times 2.46^{5/3}}{10} \right)^{3/5} = 6.47 \text{ liters}$$

$$T_2 = \frac{P_2 V_2}{nR} = \frac{10 \times 6.47}{5 \times 0.08206} = 158 \text{ K}$$

3.19 Quantitative Example Problems

- For the irreversible process, which takes the gas from the state 1 to the state 3, as $q=0$,

$$\Delta U = -w = -4000 = nc_V (T_3 - T_1) = 5 \times 1.5 \times 8.314 \times (T_3 - 300)$$

$$T_3 = 236 \text{ K}$$

- As the irreversible expansion from state 1 to state 3 was conducted adiabatically, no heat entered the system, and hence the difference between the entropy at state 3 and the entropy at state 1 is the entropy created, ΔS_{irr} , as a result of the irreversible process. This difference in entropy can be calculated by considering any reversible path from state 1 to state 3.

3.19 Quantitative Example Problems

- For a reversible constant-volume process

$$\delta q_V = nc_V dT = TdS$$

$$\text{or } dS = \frac{nc_V dT}{T}$$

$$S_a - S_1 = nc_V \ln \frac{T_a}{T_1} =$$

$$5 \times 1.5 \times 8.314 \times \ln \frac{234}{300} = -15.5 \text{ J/K}$$

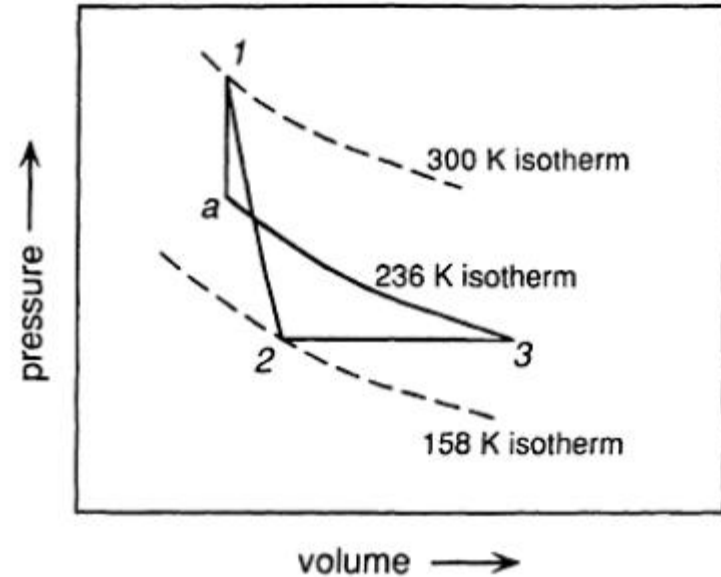


Figure 3.8 The process paths considered in Example 1.

- For the reversible isothermal expansion from state a to state 3, as $\Delta U=0$,

$$q = w = nRT \ln \frac{V_3}{V_a}$$

3.19 Quantitative Example Problems

$$V_3 = \frac{nRT_3}{P_3} = \frac{5 \times 0.08206 \times 236}{10} = 9.68 \text{ liters}$$

$$S_3 - S_a = \frac{q}{T} = nR \ln \frac{V_3}{V_a} = 5 \times 8.3144 \times \ln \frac{9.68}{2.46} = 57.0 \text{ J/K}$$

- The entropy created during the irreversible expansion is thus

$$S_3 - S_1 = -15.5 + 57.0 = 41.5 \text{ J/K}$$

- Alternatively, the state of the gas could be changed from 1 to 3 along the path 1→2→3. As the reversible adiabatic expansion from state 1 to state 2 is isentropic,

$$S_3 - S_1 = S_3 - S_2$$

3.19 Quantitative Example Problems

- for the reversible isobaric expansion from state 2 to state 3,

$$\delta q_P = nc_P dT = TdS$$

$$dS = \frac{nc_P dT}{T}$$

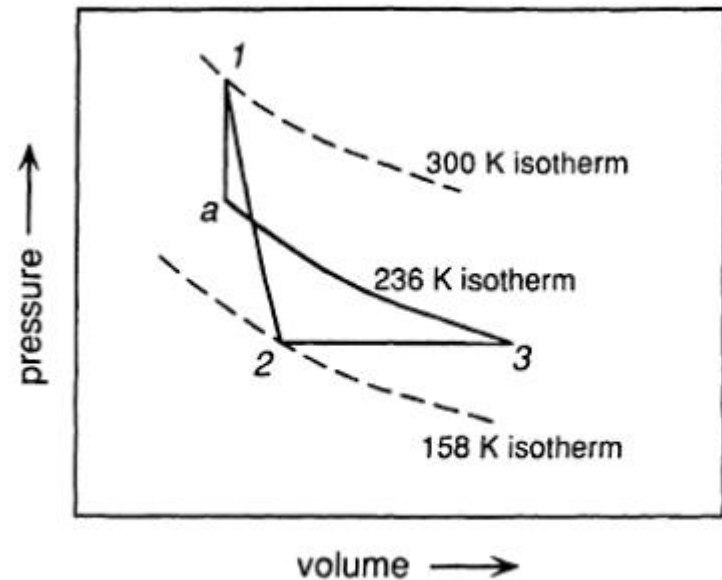


Figure 3.8 The process paths considered in Example 1.

$$S_3 - S_2 = 5 \times 2.5 \times 8.3144 \times \ln \frac{236}{158} = 41.5 \text{ J/K}$$

3.19 Quantitative Example Problems

Qualitative Problem 2

At a pressure of 1 atm, the equilibrium melting temperature of lead is 600 K, and at this temperature, the latent heat of melting of lead is 4810 J/mole. Calculate the entropy produced when 1 mole of supercooled liquid lead spontaneously freezes at 590 K and 1 atm pressure.

Given: The constant-pressure molar heat capacity of liquid lead, as a function of temperature, at 1 atm pressure is given by

$$c_p(l) = 32.4 - 3.1 \times 10^{-3} T \quad \text{J/mole} \cdot \text{K}$$

The corresponding expression for solid lead is:

$$c_p(s) = 23.56 + 9.75 \times 10^{-3} T \quad \text{J/mole} \cdot \text{K}$$

Homework

Homework

3.1 The initial state of 1 mole of a monatomic ideal gas is $P = 10$ atm and $T = 300$ K.

Calculate the change in the entropy of the gas for

- a. An isothermal decrease in the pressure to 5 atm
- b. A reversible adiabatic expansion to a pressure of 5 atm
- c. A constant-volume decrease in the pressure to 5 atm

3.3 One mole of a monatomic ideal gas undergoes a reversible expansion at constant pressure, during which the entropy of the gas increases by 14.41 J/K and the gas absorbs 6236 J of thermal energy. Calculate the initial and final temperatures of the gas. One mole of a second monatomic ideal gas undergoes a reversible isothermal expansion, during which it doubles its volume, performs 1729 J of work, and increases its entropy by 5.763 J/K. Calculate the temperature at which the expansion was conducted.

Homework

3.5 One mole of copper at a uniform temperature of 0°C is placed in thermal contact with a second mole of copper which, initially, is at a uniform temperature of 100°C. Calculate the temperature of the 2-mole system, which is contained in an adiabatic enclosure, when thermal equilibrium is attained. Why is the common uniform temperature not exactly 50°C? How much thermal energy is transferred, and how much entropy is produced by the transfer? The constant pressure molar heat capacity of solid copper varies with temperature as

$$c_p(s) = 22.64 + 6.28 \times 10^{-3} T \quad \text{J/mole} \cdot \text{K}$$

Thermodynamics of Materials & Phase transformations

The End

